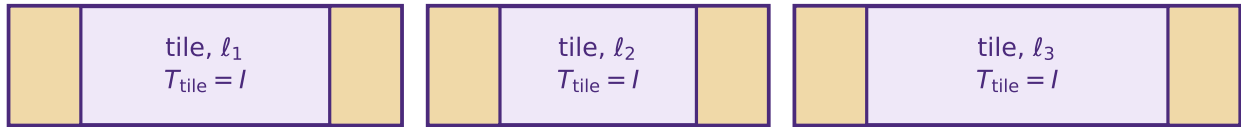


tiles of every length in $[L, 2L)$ $\Rightarrow$ every $n \geq L$ is a sum of them



$$n = l_1 + l_2 + l_3, \text{ and } T = \prod T_{\text{tile}} = I, \text{ so } \text{null } A = 2k + 2$$

shaded ends: the fixed docking pattern of $k + 1$ positions, which makes each tile's product depend on its interior alone